

CASE BRIEFING · CLASS 1

Moderna: Democratizing Artificial Intelligence

Bojinov, I., Lakhani, K., Hildebrandt, A. & Weber, J. (2025). Harvard Business School, case 625-070.

You do not need to read the case before class. We will brief it together in Session 1. This handout is for anyone who wants a head start — and as a record afterwards. The discussion questions are at the end; thinking about them in advance is optional but will make the session better.

THE SITUATION

Moderna — the mRNA company behind one of the COVID-19 vaccines — has about **6,000 employees** and is trying to win **10 new drug approvals in three years**, competing against pharmaceutical companies with over **100,000 staff**. CEO Stéphane Bancel describes it as “*a technology company that happens to do biology*.” The bet is that AI is how 6,000 people out-run 100,000.

HOW THEY GOT HERE

- **April 2023 — the forcing event.** Samsung engineers accidentally leaked source code into public ChatGPT. Moderna's exposure was worse: not just drug research, but the medical records of clinical trial volunteers. And employees were already using ChatGPT on their own.
- **May 2023 — replace, don't ban.** Public ChatGPT was cut off and replaced within two weeks by **mChat**, built on OpenAI's API so company data never trains the model.
- **Then: drive adoption.** A prompt competition (180+ entries, top prize a trip to meet OpenAI's founders), an internal AI Academy, and an eight-week training programme for senior leaders — who became the champions.
- **By December 2024 — over 1,400 custom GPTs**, almost all built by ordinary employees with no coding, not by the AI team. They eventually built a GPT called *Zeus* just to help people find the other GPTs.

THE TWO THAT MADE LEADERSHIP STOP

Self-Review GPT	DoseID
The single most-used tool in the company — 12,751 messages from 885 users. It helps employees write their own performance reviews.	Built by a physician. Recommends drug dosing for clinical trials, cutting the decision from about a month to a fortnight.
“ <i>Augmenting — and to some extent replacing — a critical process in developing employees. Maybe there are some processes that should be kept off-limits?</i> ” Tracey Franklin, Chief People & Digital Technology Officer	“ <i>Would a tool like this need regulatory approval? Clinical trials are tightly regulated.</i> ” Stéphane Bancel, CEO

One is high-stakes because of regulators and patients. The other because of people and judgment. **Neither is a technology failure** — both tools work as intended.

WHAT ELSE COULD GO WRONG

“*The risk is we have a hallucination and we send something to regulators that is incorrect. If I decentralize, I'm assuming there are enough guardrails from my quality and legal teams.*” — Stéphane Bancel

- **Four** competing travel-expense GPTs exist — and none was built by the travel-expense team. Which should an employee trust?
- **Stale tools.** Nobody knew how to retire a GPT once its underlying policy changed.
- **The counter-example is real.** Other pharmaceutical companies simply banned ChatGPT outright. Note also that Samsung's leak happened 20 days *after* the company lifted its own ban.

THE DECISION ON THE TABLE, DECEMBER 2024

1. Should Moderna take centralised ownership of GPTs? 2. Is the current governance sufficient, or does it need more oversight? 3. Does Moderna need to tell regulators that GPTs are now inside core processes?

DISCUSSION QUESTIONS — we will work these in groups

There are no wrong answers. These are about how you reason, not what you recall.

1. Moderna's most-used AI tool helps employees write their own performance reviews. You have probably used AI on a university assignment or a personal statement. Where is the line between AI helping you say what you meant, and AI replacing the thinking you were supposed to do? Is it the same line in both cases?
2. Moderna has four different travel-expense GPTs and none were built by the travel-expense team. If you were a new employee on day one, which would you use — and how would you even know which to trust?
3. One GPT recommends drug dosing for clinical trials. Another turns your photo into a cat. Both were built the same way, by employees, with no coding. What should be different about how a company treats those two — and who decides?

NOTES

1. **This is a briefing, not the case.** It summarises a copyrighted Harvard Business School case for classroom discussion. If you want the full case, including its exhibits and financials, ask about access — do not rely on this summary as a source.
2. **The case is a snapshot of December 2024.** It was published in January 2025. A great deal has changed since; part of the exercise is noticing what has and has not.
3. **This summary was generated and fact-checked using Anthropic models** against the case text. Errors are still possible — flag anything that looks wrong.